

APPLIED OPERATING SYSTEM



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

MODULE 5

MASS STORAGE MANAGEMENT



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

OBJECTIVES

Upon completion of this module, the student will be able to:

- Discuss structure of a disk;
- Explain how the system format a disk, boot from disk and read bad block disk;
- State how the system manage disk spaces;
- Define allocation methods used by the system;
- Use different disk scheduling algorithms such as **First Come First Served (FCFS)**, **Shortest Seek Time First (SSTF)**, **SCAN**, **C-SCAN**, **LOOK** and **C-LOOK** scheduling in disk scheduling problems;



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

MASS STORAGE STRUCTURE CONCEPTS



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

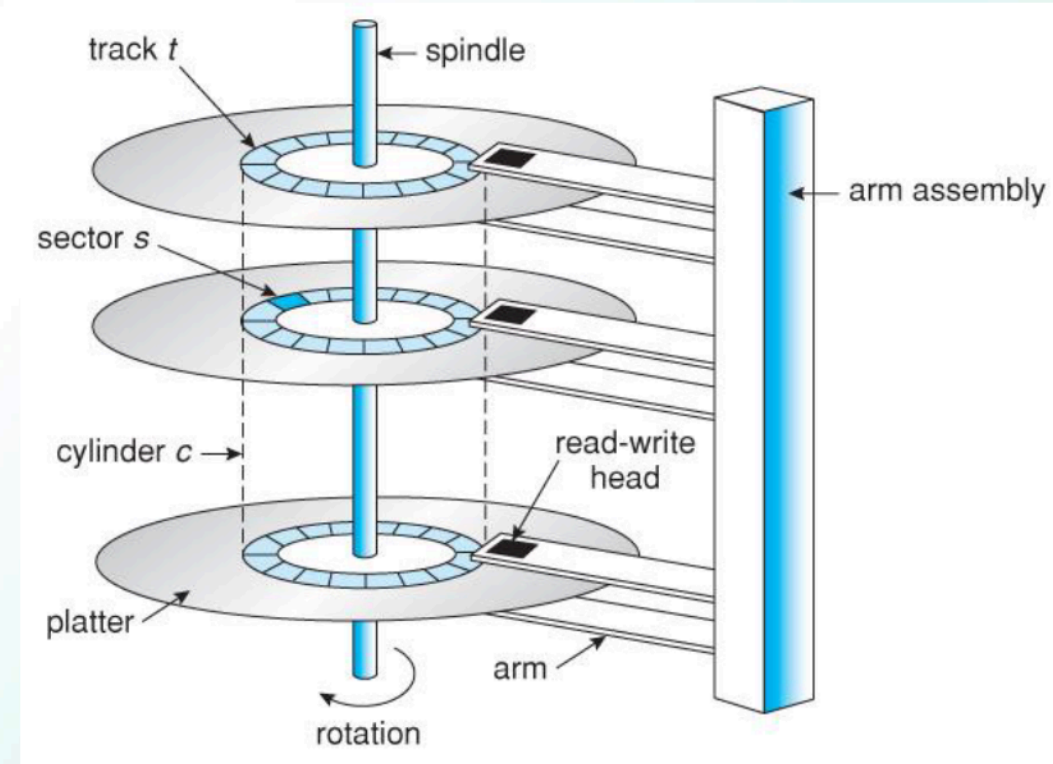
INTRODUCTION

- The main requirement of secondary storage is to be able to store a very large amount of data **permanently**.
- One of the earliest secondary-storage media is the **magnetic tape**. However, it was too slow in comparison with the access time of main memory.
- **Magnetic disks** provide the bulk of secondary storage for modern computer systems.



DISK STRUCTURE

- A magnetic disk system has several disk platters.
- Each disk platter has a flat circular shape, like a phonograph record.
- A magnetic material, similar to that of a magnetic tape or floppy diskette, covers its two surfaces.
- Information is recorded on the surfaces.



DISK STRUCTURE

- Disks are rigid **metal or glass platters** covered with magnetic recording material.
- The disk surface is logically divided into **tracks**, which are subdivided into **sectors**.
- The head does not actually touch the surface of a disk. Instead, **it floats or flies** only microns from the disk surface, supported by a cushion of air.
- Typical hard disks have a rotation speed from **4,500 to 7,200 rpm (revolution per minute)**, a 10,000 rpm drive just hit the market.
- The faster the rotation, the higher the transfer rate, but also the louder and hotter the HD.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

- You may need to cool a 7200 rpm disk with an extra fan, or its life would be much shorter. Modern HD's read all sectors of a track in one turn (Interleave 1:1). The rotation speed is **constant**.
- **Head crashes** can be problem. If the head **contacts** the disk surface (due to power failure, for example), the head will **scrape** the recording medium off the disk, destroying data that had been there.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

- **Magnetic disks** provide the bulk of secondary storage for modern computer systems.
- **Magnetic tape** was used as an early secondary-storage medium, but the access time is **much slower** than for disks.
- Magnetic tapes are currently used mainly for backup, for storage of infrequently used information, and as a medium for transferring information from one system to another.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

- The system stores information by recording it **magnetically** on the sector under the read-write head.
- A **fixed-head system** has a separate read-write head for each track. This arrangement allows the computer to switch from track to track quickly, but it requires a large number of heads, making the device extremely expensive.
- A **movable-head system** has only one read-write head per surface and the system moves the head to access a particular track (slower but less expensive). In this system, all the tracks on one drive that can be accessed without moving the heads are called a **cylinder**.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

- Floppy disks take a different approach. The disks are coated with a hard surface, so the read-write head scans it directly on the disk surface without destroying the data.
- A **3.5" floppy diskette**, which was one of the most commonly used floppy diskettes, capable of storing **1.44 MB** of data.



DISK STRUCTURE

- A disk normally has a **device directory** indicating which files are on the disk.
- The **directory lists** the file by name, and includes such information as where on the disk the file is, and what are its length, type, owner, time of creation, time of last use, protections, and so on.
- The system stores the disk directory on the device, often at some fixed disk address, such as disk address 00001.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

- Modern disks are addressed as large one-dimensional arrays of logical blocks, where the **logical block** is the smallest unit of transfer.
- The size of a logical block is usually **512 bytes**, although some disks can be low-level formatted to choose a different logical block size, such as **1,024 bytes**.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

There are two ways of reading and writing of data on disks:

1. Constant Linear Velocity (CLV)

- In disks using CLV, the density of bits (bits/unit length) per track is **uniform**.
- Since the **outer tracks are longer**, the number of bits on the outer tracks is more than inner tracks. Consequently, there are **more sectors on the outer tracks**.
- The **drive increases its rotation speed** as the head from the outer to the inner tracks **to keep the same rate of data moving the head**.
- This method is used in **CD-ROM** and **DVD-ROM drives**.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

2. Constant Angular Velocity (CAV)

- In disks using CAV, the **number of bits per track** is **uniform** (constant number of sectors).
- This means that the density of bits per track increases from outer tracks to inner tracks.
- This implies that the disk rotation stays constant.
- This method is used in **hard disks** and **floppy disks**.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK STRUCTURE

- The time it takes to access a sector depends on three parameters, the **seek time**, **latency time**, and **transfer time**.
- **Seek time** is the time it takes to move the read-write head to the correct track.
- **Latency time** is the time it takes for the sector to rotate under the head.
- **Transfer time** is the time it takes to actually transfer data between disk and main memory.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK MANAGEMENT

There are other aspects of disk management for which an operating system is responsible. They are **disk formatting, booting from disk, and bad-block recovery.**

1. Disk Formatting

- Before a computer can make use of a new disk, the disk must be broken into the sectors that the computer can understand. This process is called **physical formatting**.
- When formatted, a disk consists of a set of sectors on each track the head can address.
- Each sector has a header containing the sector number and space for error correcting code (ECC).



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK MANAGEMENT

2. Boot Block

- For a computer to start running – for instance, when it is powered up or rebooted – it needs to have an initial program to run.
- This initial program, or **bootstrap program**, tends to be simple.
- The bootstrap program initializes all aspects of the system, from CPU registers to device controllers to memory contents.

3. Bad Blocks

- Because disks have moving parts and small tolerances, they are prone to failure.
- Sometimes, the failure is complete, and the disk needs to be replaced. More frequently, one or more blocks become unreadable and not writable.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

FREE SPACE MANAGEMENT

- Since there is only a limited amount of disk space, it is necessary to reuse the space from deleted files for new files.
- To keep track of the free disk space, the system maintains a **free-space list**.
- Implementation of the free-space list:
 1. **Bit Vector**
 - A single bit represents each block.
 - If the block is **free**, the bit is **0**; otherwise, the bit is **1**.

FREE SPACE MANAGEMENT

1. Bit Vector

Example:

If blocks 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, 17, 18, 25, 26, and 27 are **free**, and the rest of the blocks are allocated, then the free-space bit map or vector would be:

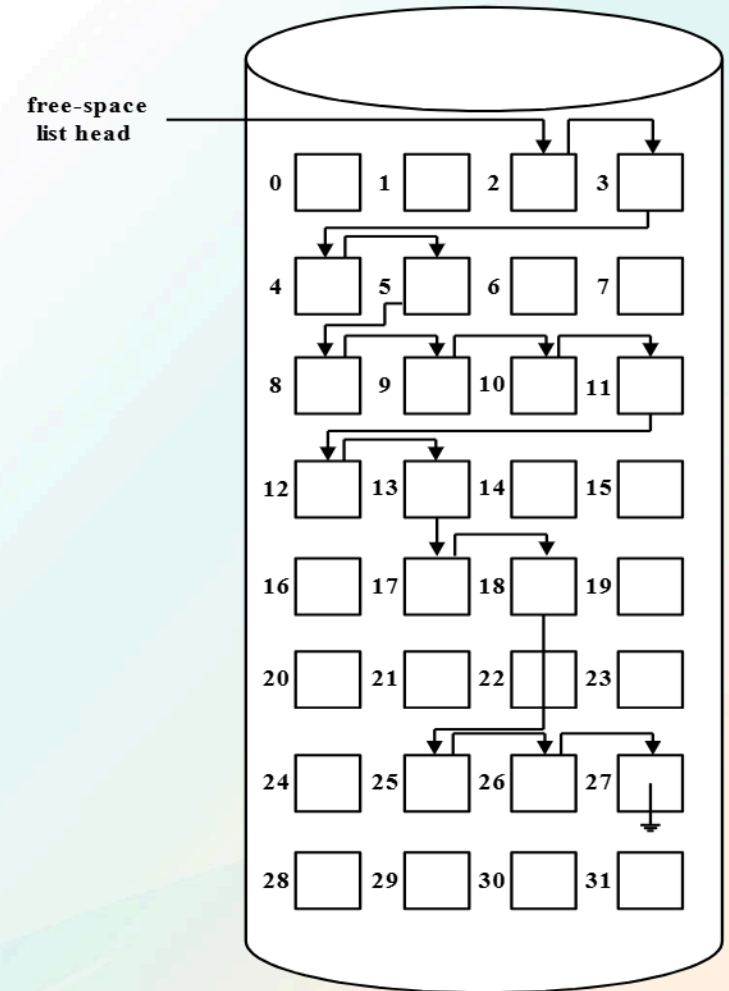
1 1 0 0 0 0 1 1 0 0 0 0 0 0 1 1 1 0 0 1 1 1 1 1 1 0 0 0 1 1 1 1....

The main disadvantage of this technique is that the OS often keeps this bit map in main memory.

FREE SPACE MANAGEMENT

2. Linked List

- Another approach is to link all the **free disk blocks together**, keeping a pointer to the first free block.
- This block contains a pointer to the next free disk block, and so on.
- This technique however, is not efficient because of the sequential nature of lists.



FREE SPACE MANAGEMENT

3. Grouping

- A modification of the free-list approach is to store the addresses of n free blocks in the first free block.
- The first $n-1$ of these are actually free.
- The last one is the disk address of another block containing the addresses of another n free blocks.
- The importance of this implementation is that the addresses of a large number of free blocks can be found quickly.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

FREE SPACE MANAGEMENT

4. Counting

- Another approach is to take advantage of the fact that, generally, several contiguous blocks may be allocated or freed simultaneously.
- Thus, rather than keeping a list of n free disk addresses, the system keeps the address of the first free block and the number n of free contiguous blocks that follow the first block.
- Each entry in the free-space list then consists of a disk address and a count.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

ALLOCATION METHODS

- There are three methods of allocating space to files so that disk space utilization is effective and access time of files is fast.

1. Contiguous Allocation

2. Linked Allocation

3. Indexed Allocation



FEU ALABANG



FEU DILIMAN



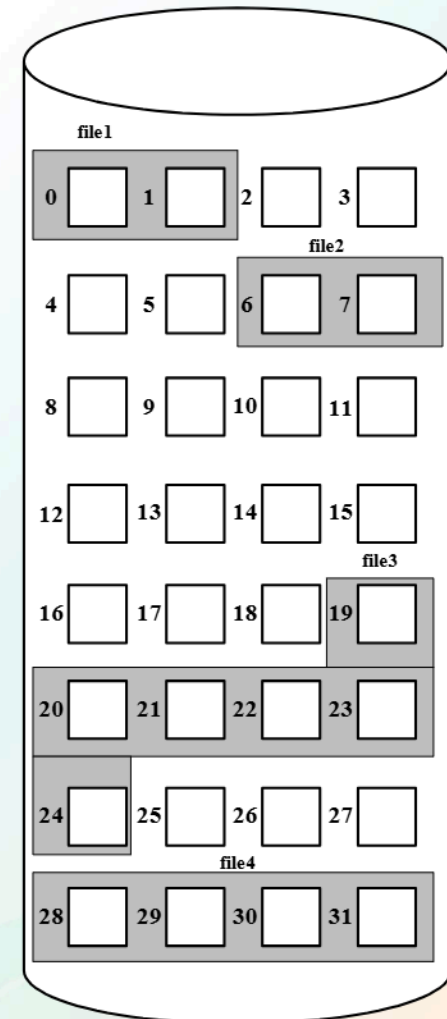
FEU TECH

Technology Driven by Innovation

ALLOCATION METHODS

1. Contiguous Allocation

- The contiguous allocation method requires **each file** to occupy a set of **contiguous addresses on the disk**.
- If the file is n blocks long, and starts at location b , then it occupies blocks $b, b + 1, b + 2, \dots, b + n - 1$.
- The directory entry for each file indicates the address of the starting block and the length of the area allocated for this file.



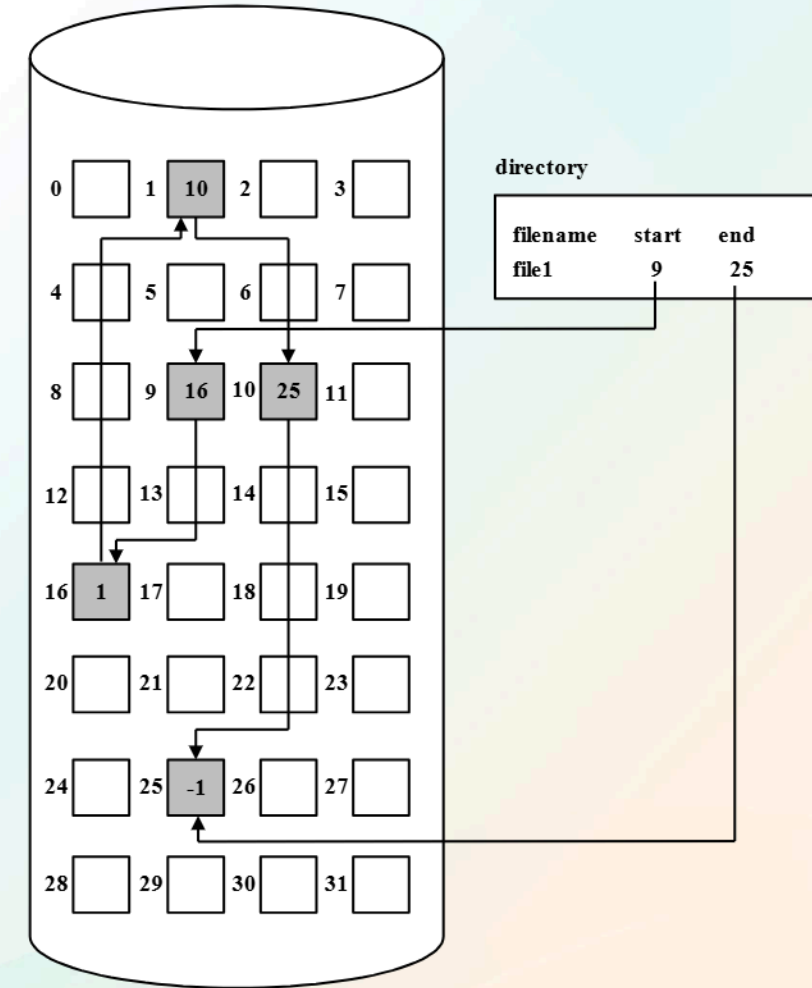
directory		
filename	start	length
file1	0	2
file2	6	2
file3	19	6
file4	28	4



ALLOCATION METHODS

2. Linked-List Allocation

- Each file is a **linked list** of disk blocks; the disk blocks may be **scattered anywhere** on the disk.
- The directory contains a pointer to the first and last blocks of the file.
- Each block contains a pointer to the next block.



ALLOCATION METHODS

- There is **no external fragmentation** with linked-list allocation.
- Any free block on the free-space list can be used to satisfy a request.
- A file can also continue to grow as long as there are free blocks.
- A disadvantage of this technique is that the pointers also occupy precious disk space.
- If a pointer requires 4 bytes out of a 512-byte block, then 0.78 % of the disk is being used for pointers, rather than information (for a 40 GB disk, that's 320 MB).



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

ALLOCATION METHODS

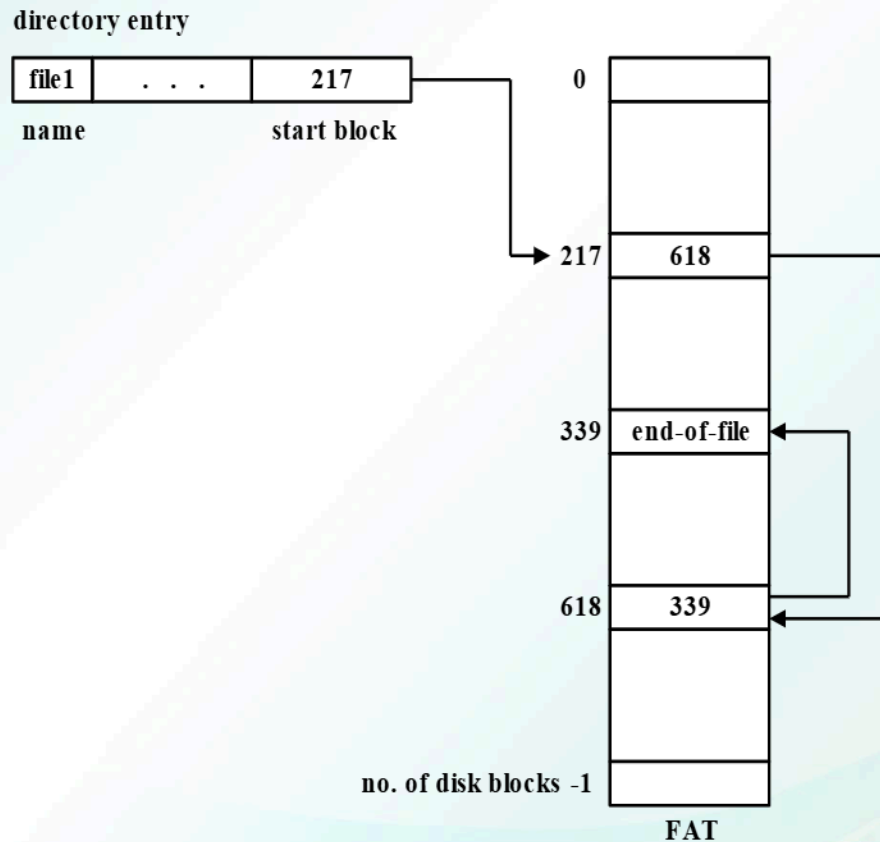
- An important variation on the linked allocation method is the use of the **file-allocation table (FAT)**.
- This table has one entry for each disk block, and is indexed by block number.
- The OS uses this table much like a linked list.
- The directory contains the block number of the first block of the file.
- The table entry indexed by that block number then contains the block number of the next block of the file. The chain continues until the last block, which has a special end-of-file value in the table entry. A zero table value indicates free memory blocks.



ALLOCATION METHODS

Example:

A file consisting of disk blocks 217, 618, and 339.



ALLOCATION METHODS

3. Indexed Allocation

- Indexed allocation is similar to linked allocation except for the fact that the pointers to the blocks are not scattered but are grouped in one location which is the **index block**.
- Each file has its own index block, which is an array of disk-block addresses. The *ith* entry in the index block points to the *ith* block of the file.
- The directory contains the address of the index block.



FEU ALABANG



FEU DILIMAN

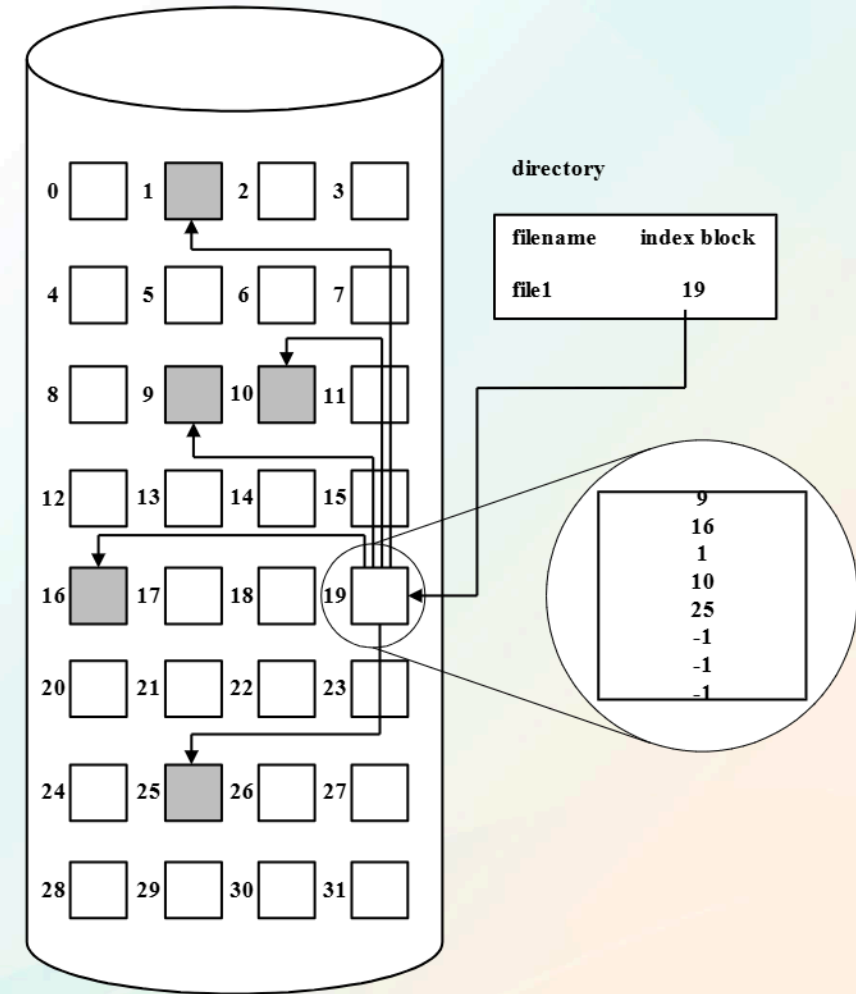


FEU TECH

Technology Driven by Innovation

ALLOCATION METHODS

- Indexed allocation supports direct access, without suffering from external fragmentation.
- However, it suffers from wasted space. The pointer overhead of the index block is generally greater than the pointer overhead of linked allocation.
- The OS must allocate an entire index block, even if only one or two pointers will be non-nil.



DISK SCHEDULING TECHNIQUES



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK SCHEDULING

- Since most jobs depend heavily on the disk for loading and input and output files, it is important that disk service be as fast as possible.
- The OS can **improve** on the average disk service time by **scheduling the requests for disk access**.
- If the desired disk drive and controller are available, it can service the request immediately. However, while the drive or controller is serving one request, any additional requests, normally from other processes, should go to a queue.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK SCHEDULING

- **Disk Scheduling Algorithms**

1. FCFS Scheduling
2. SSTF Scheduling
3. SCAN Scheduling
4. C-SCAN Scheduling
5. LOOK Scheduling
6. C-LOOK Scheduling



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

FCFS DISK SCHEDULING

1. FCFS Scheduling

- The simplest form of scheduling is the **first-come first-served** (FCFS).

Example:

Assume that there are 200 tracks (0-199) and the **read-write head** is currently at **track 53**. Consider now an ordered disk queue with requests involving tracks.

98, 183, 37, 122, 14, 124, 65, and 67



FEU ALABANG



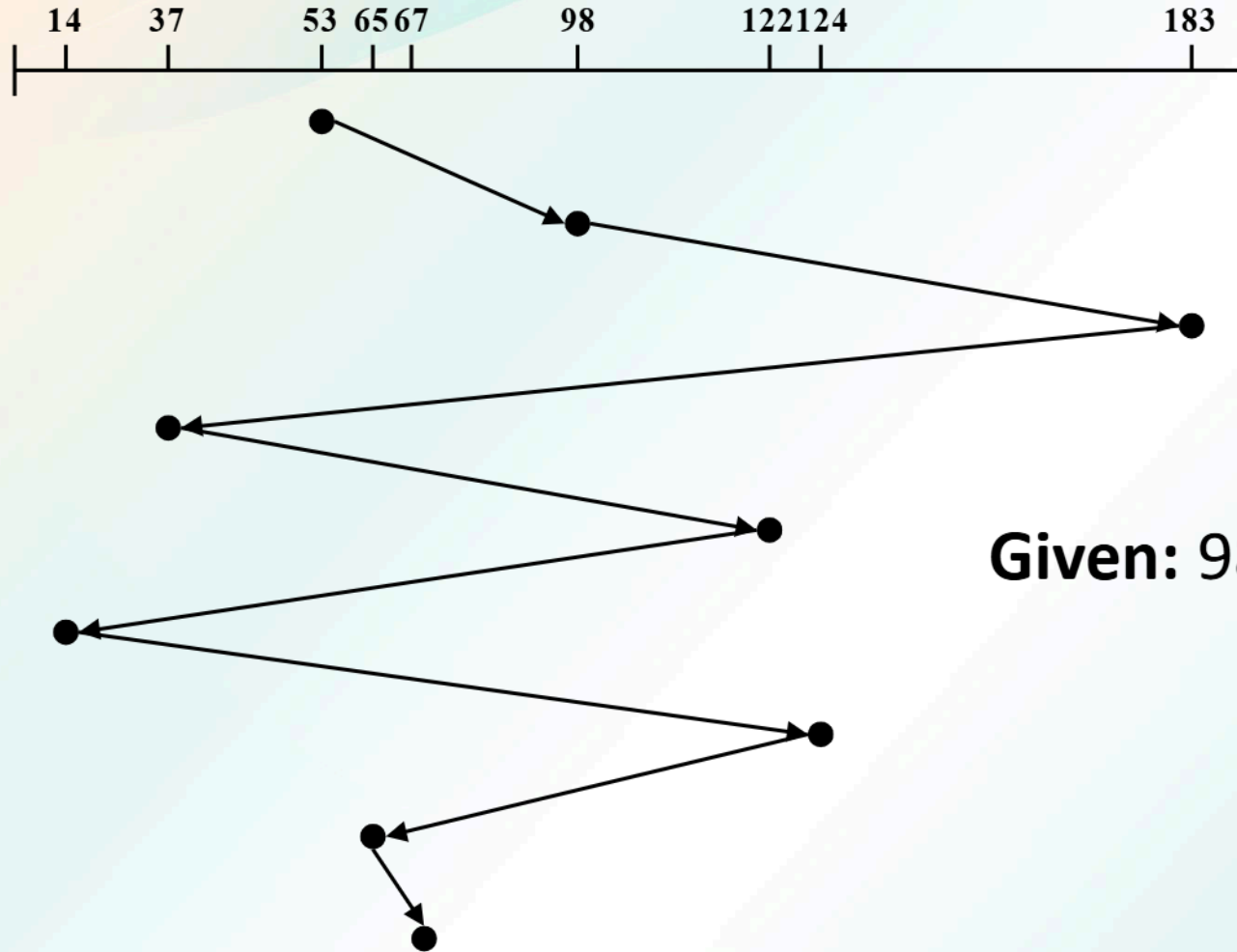
FEU DILIMAN



FEU TECH

Technology Driven by Innovation

FCFS DISK SCHEDULING



Given: 98, 183, 37, 122, 14, 124, 65, and 67

FCFS DISK SCHEDULING

Computing for the **Total Head Movement (THM)**:

from 53 to 98	=	98 - 53	=	45
from 98 to 183	=	183 - 98	=	85
from 183 to 37	=	183 - 37	=	146
from 37 to 122	=	122 - 37	=	85
from 122 to 14	=	122 - 14	=	108
from 14 to 124	=	124 - 14	=	110
from 124 to 65	=	124 - 65	=	59
from 65 to 67	=	67 - 65	=	2
Total Head Movement			=	640 tracks



SSTF DISK SCHEDULING

2. SSTF Scheduling

- The **Shortest Seek Time First** selects the request with the minimum seek time from the current head position. This means that the system moves the head to the closest track in the request queue.

Example:

Assume that the read-write head is currently at **track 53**. Consider now an ordered disk queue with requests involving track.

98, 183, 37, 122, 14, 124, 65, and 67



FEU ALABANG



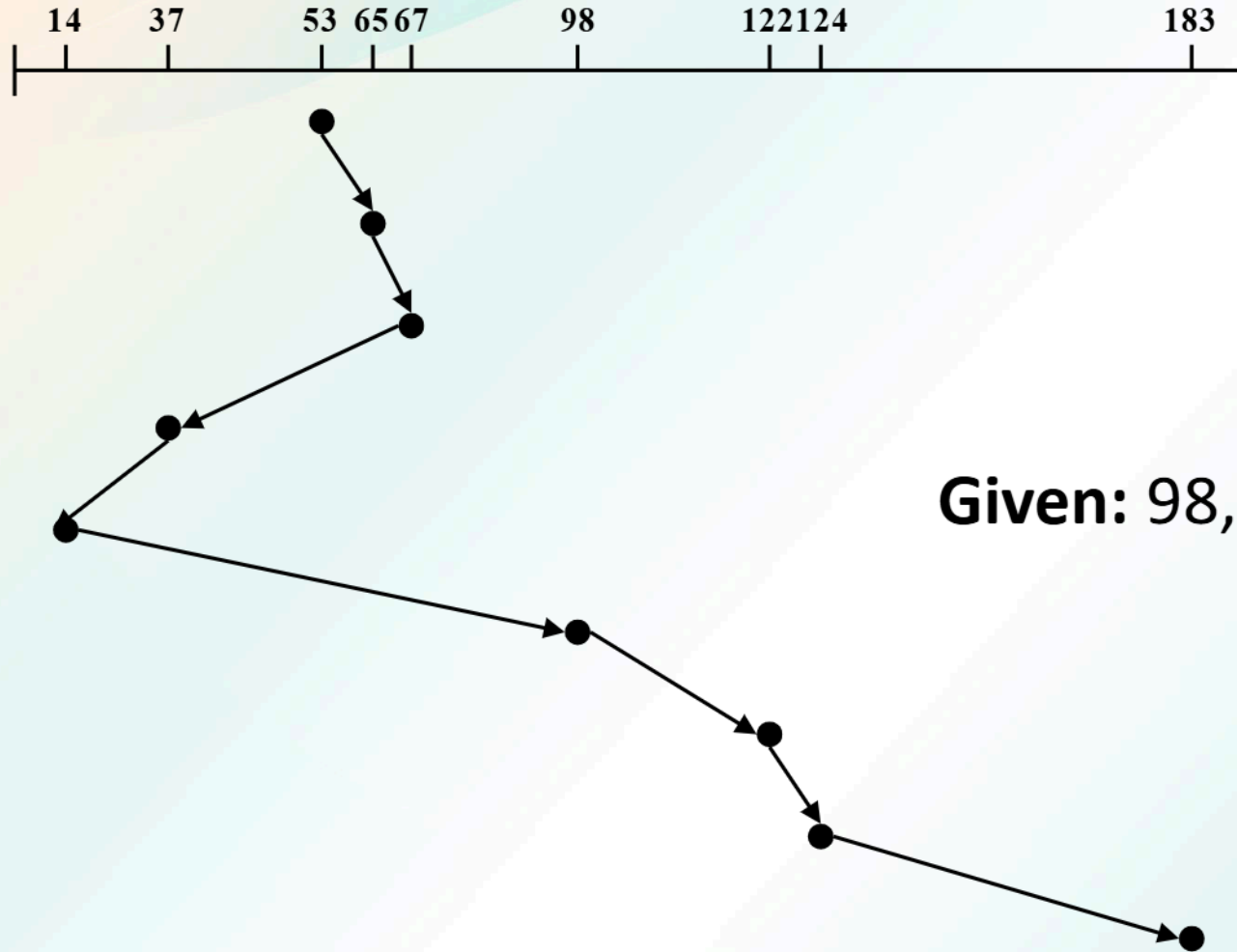
FEU DILIMAN



FEU TECH

Technology Driven by Innovation

SSTF DISK SCHEDULING



Given: 98, 183, 37, 122, 14, 124, 65, and 67



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

SSTF DISK SCHEDULING

Computing for the **Total Head Movement (THM)**:

from 53 to 65	=	65 - 53	=	12
from 65 to 67	=	67 - 65	=	2
from 67 to 37	=	67 - 37	=	30
from 37 to 14	=	37 - 14	=	23
from 14 to 98	=	98 - 14	=	84
from 98 to 122	=	122 - 98	=	24
from 122 to 124	=	124 - 122	=	2
from 124 to 183	=	183 - 124	=	59
Total Head Movement			=	236 tracks

SSTF DISK SCHEDULING

- This scheduling method results in a total head movement of 236 tracks. This algorithm would result in a substantial improvement in average disk service.
- The problem with SSTF scheduling is that it may cause **starvation** of some requests.
- In theory, a continual stream of requests near one another could arrive, causing the request for a farther track to wait indefinitely.
- The SSTF algorithm, although a substantial improvement over the FCFS algorithm, is not optimal.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

SCAN DISK SCHEDULING

3. SCAN Scheduling

- The read-write head starts at one end of the disk, and moves toward the other end, servicing requests as it reaches each track, until it gets to the other end of the disk.
- At the other end, the direction of head movement reverses and servicing continues.
- The head continuously scans the disk from end to end.
- The SCAN algorithm is sometimes called the **elevator algorithm**, since it is similar to the behavior of elevators as they service requests to move from floor to floor in a building.



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

SCAN DISK SCHEDULING

Example:

Assume that there are 200 tracks (0-199) and the read-write head is currently at track 53 (**going to the direction of track 0**). Consider now an ordered disk queue with requests involving tracks:

98, 183, 37, 122, 14, 124, 65, and 67.



FEU ALABANG



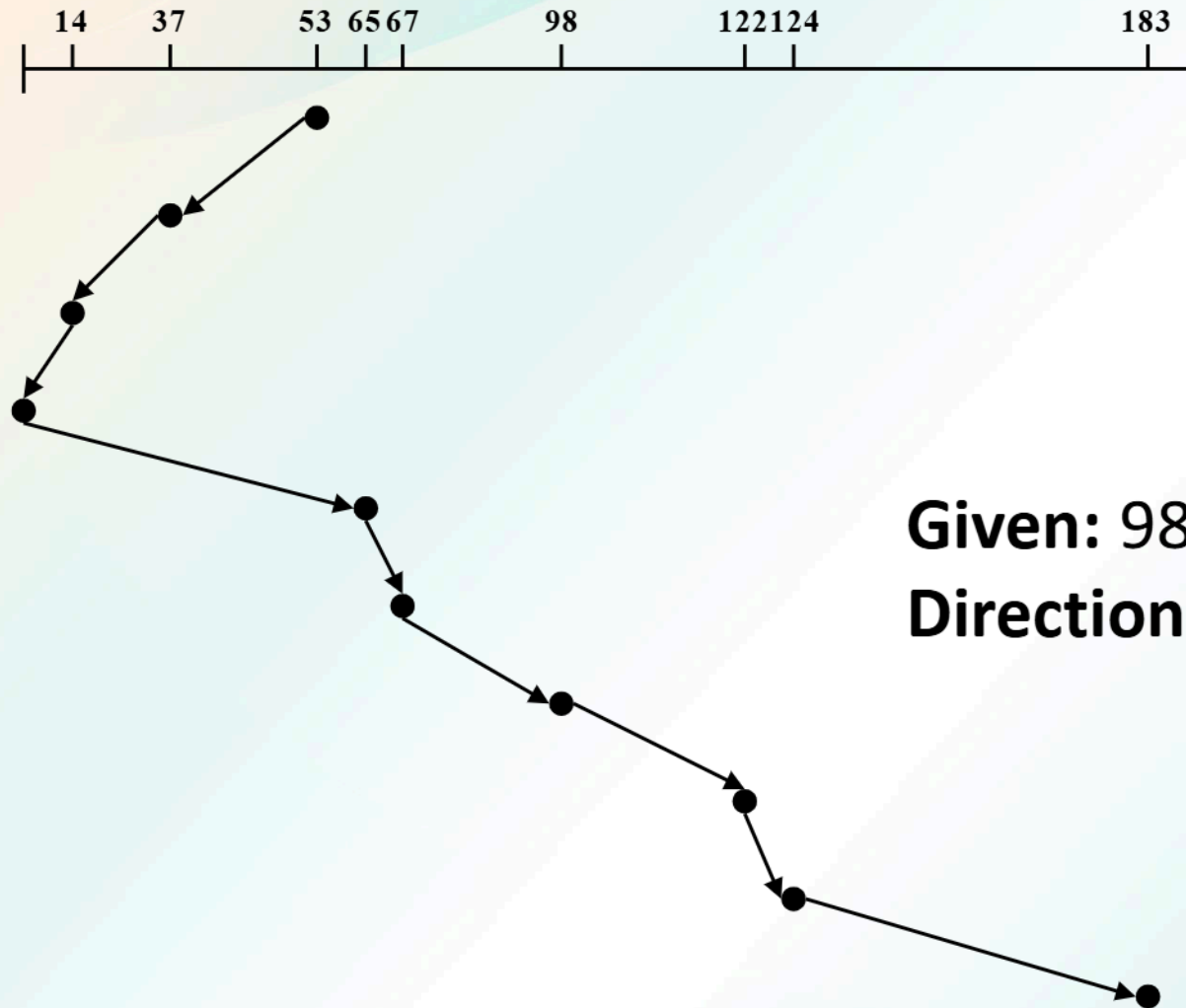
FEU DILIMAN



FEU TECH

Technology Driven by Innovation

SCAN DISK SCHEDULING



Given: 98, 183, 37, 122, 14, 124, 65, and 67

Direction: Towards track 0 direction

SCAN DISK SCHEDULING

Computing for the **Total Head Movement (THM)**:

from 53 to 37	=	53 - 37	=	16
from 37 to 14	=	37 - 14	=	23
from 14 to 0	=	14 - 0	=	14
from 0 to 65	=	65 - 0	=	65
from 65 to 67	=	67 - 65	=	2
from 67 to 98	=	98 - 67	=	31
from 98 to 122	=	122 - 98	=	24
from 122 to 124	=	124 - 122	=	2
from 124 to 183	=	183 - 124	=	59
Total Head Movement			=	236 tracks



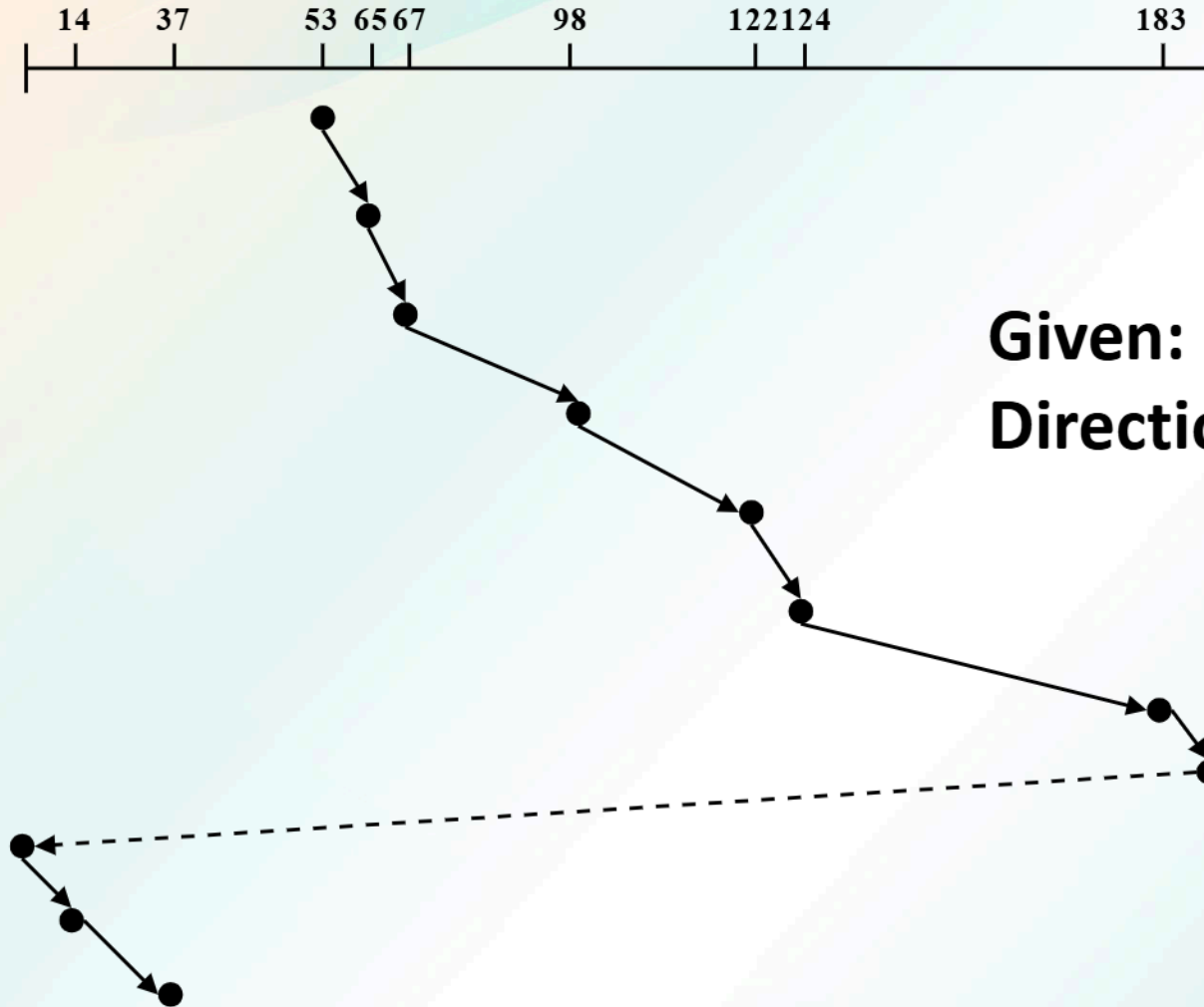
C-SCAN DISK SCHEDULING

4. C-SCAN Scheduling

- The **Circular-SCAN** algorithm is a variant of the SCAN algorithm.
- As does SCAN scheduling, C-SCAN scheduling moves the head from one end of the disk to the other, servicing requests as it goes.
- When it **reaches the other end**, however, it **immediately returns to the beginning** of the disk without servicing any requests on the return trip.
- C-SCAN scheduling essentially treats the disk as though it were circular, with the last track adjacent to the first one.



C-SCAN DISK SCHEDULING



Given: 98, 183, 37, 122, 14, 124, 65, and 67

Direction: Towards upward direction

C-SCAN DISK SCHEDULING

Computing for the **Total Head Movement (THM)**:

from 53 to 65	=	65 - 53	=	12
from 65 to 67	=	67 - 65	=	2
from 67 to 98	=	98 - 67	=	31
from 98 to 122	=	122 - 98	=	24
from 122 to 124	=	124 - 122	=	2
from 124 to 183	=	183 - 124	=	59
from 183 to 199	=	183 - 199	=	16
from 199 to 0	=	199 - 0	=	199
from 0 to 14	=	14 - 0	=	14
from 14 to 37	=	37 - 14	=	23

Total Head Movement = **382 tracks**



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

LOOK DISK SCHEDULING

5. LOOK Scheduling

- In SCAN and C-SCAN scheduling, the head always moves from one end of the disk to the other.
- More commonly, the head is only moved as far as the last request in each direction.
- As soon as there are no requests in the current direction, the head movement is reversed.
- These versions of SCAN and C-SCAN scheduling are called **LOOK** (“look” for a request before moving in that direction) and **C-LOOK** scheduling.



FEU ALABANG



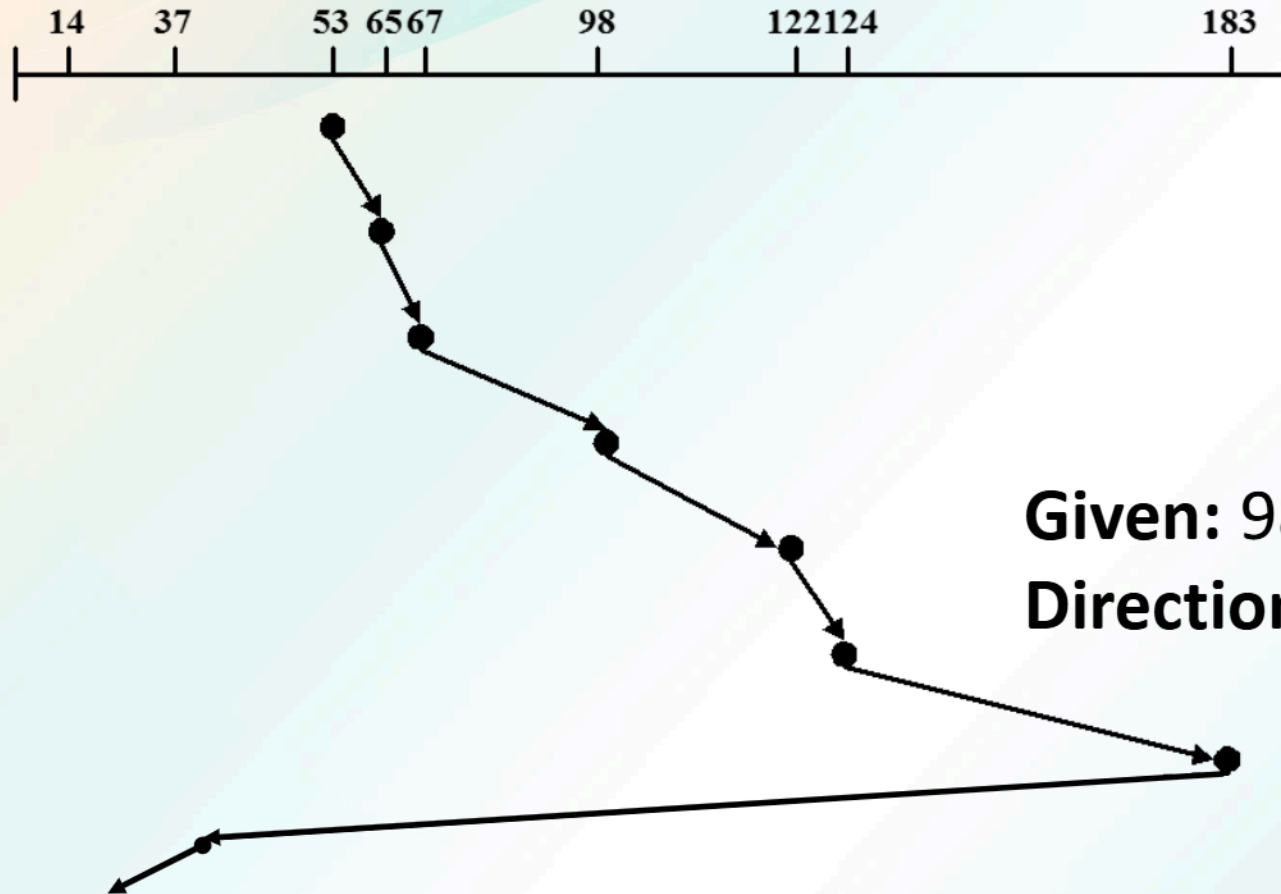
FEU DILIMAN



FEU TECH

Technology Driven by Innovation

LOOK DISK SCHEDULING



Given: 98, 183, 37, 122, 14, 124, 65, and 67
Direction: Towards upward direction

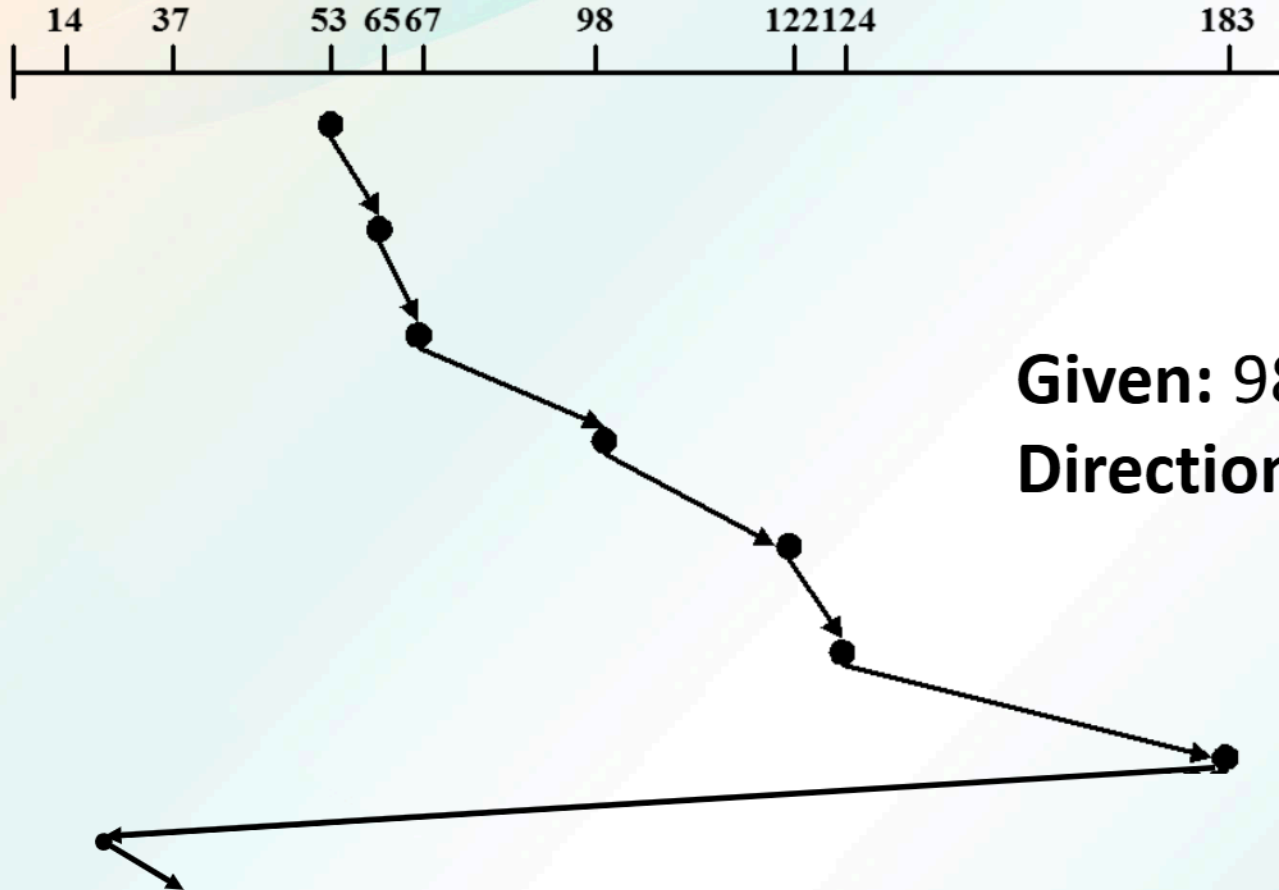
LOOK DISK SCHEDULING

Computing for the **Total Head Movement (THM)**:

from 53 to 65	=	65 - 53	=	12
from 65 to 67	=	67 - 65	=	2
from 67 to 98	=	98 - 67	=	31
from 98 to 122	=	122 - 98	=	24
from 122 to 124	=	124 - 122	=	2
from 124 to 183	=	183 - 124	=	59
from 183 to 14	=	183 - 37	=	146
from 14 to 37	=	37 - 14	=	23
Total Head Movement			=	299 tracks



C-LOOK DISK SCHEDULING



Given: 98, 183, 37, 122, 14, 124, 65, and 67
Direction: Towards upward direction

C-LOOK DISK SCHEDULING

Computing for the **Total Head Movement (THM)**:

from 53 to 65	=	65 - 53	=	12
from 65 to 67	=	67 - 65	=	2
from 67 to 98	=	98 - 67	=	31
from 98 to 122	=	122 - 98	=	24
from 122 to 124	=	124 - 122	=	2
from 124 to 183	=	183 - 124	=	59
from 183 to 14	=	183 - 14	=	169
from 14 to 37	=	37 - 14	=	23
Total Head Movement			=	322 tracks



SELECTING DISK SCHEDULING ALGORITHM

- **SCAN** and **C-SCAN** are more appropriate for systems that place a heavy load on the disk.
- If the queue seldom has more than one outstanding request, then all scheduling algorithms are effectively equivalent. In this case, FCFS is a reasonable algorithm (due to its simplicity).



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK MANAGEMENT

There are other aspects of disk management for which an operating system is responsible. They are **disk formatting, booting from disk, and bad-block recovery.**

1. Disk Formatting

- Before a computer can make use of a new disk, the disk must be broken into the sectors that the computer can understand. This process is called **physical formatting**.
- When formatted, a disk consists of a set of sectors on each track the head can address.
- Each sector has a header containing the sector number and space for error correcting code (ECC).



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

DISK MANAGEMENT

2. Boot Block

- For a computer to start running – for instance, when it is powered up or rebooted – it needs to have an initial program to run.
- This initial program, or **bootstrap program**, tends to be simple.
- The bootstrap program initializes all aspects of the system, from CPU registers to device controllers to memory contents.

3. Bad Blocks

- Because disks have moving parts and small tolerances, they are prone to failure.
- Sometimes, the failure is complete, and the disk needs to be replaced. More frequently, one or more blocks become unreadable and not writable.



REFERENCES

- **Siberschatz, A. (2018). Operating System Concepts, Wiley.**
- **Tomsho, G. (2019). Guide to Operating Systems, Cengage Learning.**



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation